UNIVERSITY OF UYO

DEPARTMENT OF MATHEMATICS

2021/2022 FIRST SEMESTER EXAMINATION

COURSE CODE: MTH212: COURSE TITLE: INTRODUCTORY ALGEBRA

Time: 2 HRS

Instructions: Answer Three Questions

(1) (a) Define: (i) groupoid; (ii) Monoid; (iii) semigroup; (iv) Group and (v) Abelian group

(b) Given that $X = \{1, -1, i, -i\} \subset \mathbb{C}$ where $i = \sqrt{-1}$. Show that $(X, .)$ forms a group where . denotes the usual binary operation of multiplication.

(2) (a) Define the followings with examples: (i) associativity and commutativity; (ii) right-distributivity and left-distributivity; (iii) commutative ring and ring with identity; (iv) divisors of zero and (v) illustrate two examples of rings with no zero divisors

(b) Let $X = \{1, -1, i, -i\} \subset \mathbb{C}$ where $i = \sqrt{-1}$ and define addition $+$ and multiplication $\times$ on X by the tables below:

$+$	0	1	-1	i	$-i$
0	0	1	-1	i	$-i$
1	1	1	0	i	$-i$
-1	-1	0	-1	i	$-i$
i	i	i	i	i	0
$-i$	$-i$	$-i$	$-i$	0	$-i$

$\times$	0	1	-1	i	$-i$
0	0	0	0	0	0
1	0	1	-1	i	$-i$
-1	0	-1	1	$-i$	i
i	0	i	$-i$	-1	1
$-i$	0	$-i$	i	1	-1

Comment on ring properties of $(X, +, \times)$ and identify the zero and identity elements of the ring.

(3) (a) Draw up a truth table for each of the following propositional forms: (i) $(r \wedge P) \vee (\sim q \wedge \sim p)$ (ii) $(p \longrightarrow r) \wedge (r \longrightarrow q)$ where p are T, T, T, T, F, F, F, F, q are T, T, F, F, T, T, F, F and r are T, F, T, F, T, F, T, F. Note that $T = True$ and $F = False$.

(b) (i) Define the concepts of addition on natural numbers and state the cancellation law of the addition; (ii) define the concept of multiplication of natural numbers and state cancellation and distributive laws of the multiplication.

(c) Let $\mathbb{Z}$ be the set of integers and define the operation $\circledast$ by $x \circledast y = x + y - xy$. Find (i) whether $\circledast$ is commutation, (ii) the identity (iii) the inverse of x.

(4) (a) Define the following terms with examples: (i) A binary relation (ii) Eqivalence relation (iii) Partial order (iv) totaly ordered (v) well-ordered.

(b) Prove that (i) $(A \cup B)' = A' \cap B'$, (ii) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

(5) (a) State the Peano postulates and define the concept of natural numbers

(b) (i) Define the binary relation of *equivalence(modulo n)* with examples (ii) Prove that the relation $a \equiv b(modn)$ is an equivalence